SOMAIYA VIDYAVIHAR UNIVERSITY

28-05-2025 (E)

Semester: Jan 2025 to April-2025		
Maximum Marks: 50 Examination: End-Semester Examination		Duration: 2 Hrs
Programme code: 01 Programme: Computer Engineering	Class: S.Y	Semester: III (SVU 2023)
Institute: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 216U01C303	Name of the Course: Object Oriented Programming Methodology	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Explain the concept of a Vector in object-oriented programming. How do vectors differ from arrays, and what are their advantages in terms of memory management and dynamic resizing?	05
ii)	Write a program that uses a for-each loop to print the highest marks in an array storing marks of 5 students.	05
iii)	Distinguish between Abstract Class and Interface.	05
Q.2	Attempt any **one**	10
i)	Create an interface Shape with methods calculateArea() and displayInfo(). Implement the interface in two classes: Circle and Rectangle. Provide methods to set dimensions and calculate areas for each shape. Implement the displayInfo() method to display shape information. Now, create an interface CloneableShape that extends the Shape interface and includes a method cloneShape(). Implement this interface in the Circle and Rectangle classes. Implement the cloneShape() method to create a deep copy of the shape. In the Main class, create instances of both shapes, clone a shape, and display the information of the original shape and the cloned shape.	10
ii)	Write a program to demonstrate the concept of constructors in an abstract class. Your program should include the following: a. Create an abstract class Animal with a constructor that accepts the animal's name and prints a message indicating the constructor's execution. o The abstract class should also declare an abstract method sound() that must be implemented by its subclasses. b. Create two concrete subclasses: o Dog and Cat, each extending the abstract class Animal. o In each subclass, define a constructor that calls the superclass constructor using super(). Print a message indicating the subclass constructor's execution. o Implement the abstract sound() method in each subclass. For a dog, the method should print a message that the dog barks. For a cat, the method should print a message that the cat meows.	10

	c. In the main class, instantiate objects of the Dog and Cat classes with appropriate names (e.g., "Buddy" for the dog and "Whiskers" for the cat). ○ Call the sound() method on both objects and ensure the output demonstrates the correct order of constructor calls and method execution.	
Q.3	Attempt any **one**	**10**
i)	A college has four sports namely RED, YELLOW, BLUE and GREEN. The house is allotted on the basis of roll number. The roll number is divided by 4 and if the remainder is 0 then the house is RED and so on till for GREEN house is allotted. Write a program to accept the roll number and display the house. Program should generate an User defined Exception.	**10**
ii)	Write a program which accepts marks of a student (between 0 to 100) and checks whether it is within the range or not. If it is within the range then it displays "marks entered successfully", if not then it throws the exception of user defined class "MarksOutOfRangeException". The class should contain an appropriate toString method to describe the object the class with the out of range marks entered by the user.	**10**
Q.4	Attempt the following	**10**
	Explain Thread life cycle. Develop a Java program that simulates a simple digital clock. Implement a thread that updates and displays the current time every second. Utilize the ***sleep*** method to control the timing of the updates.	**10**
Q.5	Attempt the following	**10**
	i. Explain the uses of the keyword 'super' with examples. ii. Explain the 'synchronised' keyword in multithreading.	**5+5**

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